

RESEARCH LETTER

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Key Points:

- New spaceborne lidar CATS is used to investigate the distance between the smoke layer and underlying clouds in SE Atlantic region
- Smoke layer is found to be much closer to the underlying clouds than previously expected based on CALIPSO 532 nm observations
- This suggests a potentially important microphysical indirect effects of smoker aerosols through cloud top entrainment

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Seasonally transported aerosol layers over southeast Atlantic are closer to underlying clouds than previously reported

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Abstract From June to October, low-level clouds in the southeast (SE) Atlantic often underlie seasonal aerosol layers transported from African continent. Previously, the Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO) 532 nm lidar observations have been used to estimate the relative vertical location of the above-cloud aerosols (ACA) to the underlying clouds. Here we show new observations from NASA's Cloud-Aerosol Transport System (CATS) lidar. Two seasons of CATS 1064 nm observations reveal that the bottom of the ACA layer is much lower than previously estimated based on CALIPSO 532 nm observations. For about 60% of CATS nighttime ACA scenes, the aerosol layer base is within 360 m distance to the top of the underlying cloud. Our results are important for future studies of the microphysical indirect and semidirect effects of ACA in the SE Atlantic region.

1. Introduction

Every year from about June to October over the southeast (SE) Atlantic, the prevailing easterly winds in the free troposphere often transport the smoke and pollution aerosols from the African continent to the west, over the ocean where extensive marine boundary layer (MBL) clouds persist for most of the year [Adebiyi and Zuidema, 2016]. This leads to a near-persistent seasonal biomass burning aerosol layer over MBL clouds in SE Atlantic [Devasthale and Thomas, 2011; Zhang et al., 2016].

As summarized in Yu and Zhang [2013] instruments onboard NASA's A-Train satellite constellation provide valuable observations of the aerosol layer and underlying clouds. In particular, the lidar on the spaceborne mission CALIPSO provides unique observations of the vertical distribution of the aerosol layer that have been widely used to characterize the aerosol layer above cloud over SE Atlantic [Chand et al., 2008; Yu et al., 2010; Devasthale and Thomas, 2011; Meyer et al., 2013] and assess its impacts on the radiation budget [Chand et al., 2009; Zhang et al., 2016].

The seasonally transported SE Atlantic aerosol layer can influence the regional radiative energy budget through the direct radiative effect (DRE) [Chand et al., 2009; Zhang et al., 2016]. The absorption by aerosol layer can also influence the thermodynamical structure of lower atmosphere and in turn change cloud field, which is known as the semidirect effect [Johnson et al., 2004; Wilcox, 2010; Sakaeda et al., 2011; Wilcox, 2012]. The sign and magnitude of the semidirect effect are strongly dependent on the vertical distribution of aerosol with respect to the underlying clouds [Johnson et al., 2004]. In addition to DRE and semidirect effect, the aerosol particles could be entrained into the clouds and activated as cloud condensation nuclei, giving rise to the so-called aerosol indirect effects [Costantino and Bréon, 2010, 2013; Painemal et al., 2015]. Intuitively, the closer the bottom of the aerosol layer gets to the top of underlying cloud, the more likely the aerosol particles are entrained into the cloud. Previous studies have used the 532 nm observations from the CALIPSO lidar to estimate the distance from the aerosol layer bottom to the cloud top (referred to hereafter as AB2CT distance for short). Costantino and Bréon [2010] show that 84% of the time the AB2CT distance in SE Atlantic is larger than 250 m. Devasthale and Thomas [2011] found that in 0° to 30°S region, 90–95% of above-cloud-aerosol cases have an AB2CT distance greater than 100 m. Yu et al. [2010] derived the average AB2CT of 1700 m over a 2 year period in SE Atlantic. These analyses based on Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP) 532 nm observations seem to indicate that the seasonal aerosol layer in SE Atlantic is well separated from the underlying

clouds, and thus, the aerosol indirect effects may be secondary in comparison to the aerosol direct and semidirect effects [e.g., *Sakaeda et al.*, 2011].

It is known that the CALIOP 532 nm based layer detection often misses the lowest boundary of a thick aerosol layer, thereby biasing the bottom of the aerosol layer too high. This may be especially problematic for daytime observations [*Meyer et al.*, 2013]. Recently, several novel remote sensing techniques have been developed to retrieve the AOD (aerosol optical depth) of above-cloud-absorbing aerosol layers from passive sensors [e.g., *Waquet et al.*, 2009; *Torres et al.*, 2011; *Meyer et al.*, 2015]. In addition, an alternative lidar method has been developed for CALIOP, utilizing signals from the underlying cloud instead of the attenuated backscatter profile [*Hu et al.*, 2007; *Liu et al.*, 2015]. When compared with the retrievals from passive sensors and the alternative CALIOP algorithm, the operational 532 nm CALIOP AOD retrievals are systematically biased low by 26% on average [*Liu et al.*, 2015] and can be up to a factor of 5 lower [*Jethva et al.*, 2014]. A likely explanation for this bias is that the strong aerosol attenuation at 532 nm by the upper portion of the aerosol layer together with the small backscatter cross section of the aerosol particles substantially weakens the attenuated backscatter signal from the lower part of the aerosol layer to a level under the detection threshold of CALIOP [*Kacenelenbogen et al.*, 2011; *Torres et al.*, 2013; *Jethva et al.*, 2014; *Liu et al.*, 2015]. This laser attenuation issue leads to an overestimation of the aerosol layer bottom height (too high), an underestimation of the physical thickness of the aerosol layer (too thin), and thereby an underestimation of AOD (too small).

In this study, we seek to shed new light on the vertical distribution of the SE Atlantic absorbing aerosol layer with respect to the underlying clouds using observations from NASA's Cloud-Aerosol Transport System (CATS) mission. Because of instrument and algorithm differences, CATS ACA retrieval suffers much less from the laser saturation-induced bias than CALIOP 532 nm algorithm. We do a comparative analysis of CATS and CALIOP retrievals in the SE Atlantic region for two recent biomass burning seasons (2015 and 2016). As shown in the latter, the CATS 1064 nm observations suggest that the bottom of the ACA layer is much lower, and therefore closer to underlying cloud top, than previously estimated based on CALIOP 532 nm observations. Our results are important for future studies of the microphysical indirect, as well as the semidirect, effects of ACA on underlying clouds.

2. Data

The occurrence frequency of above-cloud-aerosol in the SE Atlantic (20°W to 20°E; 30°S to 10°N) is highest during July-to-October (JASO) with the peak during August–September [*Zhang et al.*, 2016]. In this study, we focus on the two biomass burning seasons (JASO) of 2015 and 2016 so that we can directly compare CALIPSO and CATS (Figure 1).

2.1. CALIOP

The lidar instrument onboard the CALIPSO mission, which has an orbital height of ~700 km, is the Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP). CALIOP directly measures the range-resolved total (particulate plus molecular) attenuated backscatter signal at two wavelengths, 532 nm and 1064 nm, using analog detection. In addition to the total attenuated backscatter, CALIOP also measures two orthogonal polarized components of the 532 nm backscatter signal [*Winker et al.*, 2009]. The accuracy of the CALIOP Level-2 (L2) data products (aerosol type, particulate backscatter and extinction coefficient, optical depth) is dependent on the accurate detection of cloud and aerosol layers.

Uniform cloud and aerosol layer detection and cloud-aerosol discrimination (CAD) techniques are challenging due to the complexity of atmospheric scenes encountered. The current version CALIOP selective, iterated boundary location (SIBYL) algorithm uses the 532 nm total attenuated backscattered signals to determine boundaries of cloud and aerosol layers, with a typical vertical resolution of 30 m [*Vaughan et al.*, 2009]. The SIBYL scheme detects atmospheric features by iteratively comparing horizontally averaged CALIOP 532 nm total attenuated backscatter profiles at multiple horizontal resolutions. The CALIOP CAD algorithm is a multidimensional probability distribution function (PDF) technique [*Liu et al.*, 2004, 2009] based on statistical differences of several cloud and aerosol properties (e.g., layer-integrated 532 nm attenuated backscatter and layer-integrated backscatter color ratio). Previous studies have shown that the SIBYL and CAD algorithms perform well for cirrus clouds and several aerosol types [*McGill et al.*, 2007; *Yorks et al.*, 2011; *Burton et al.*, 2013].

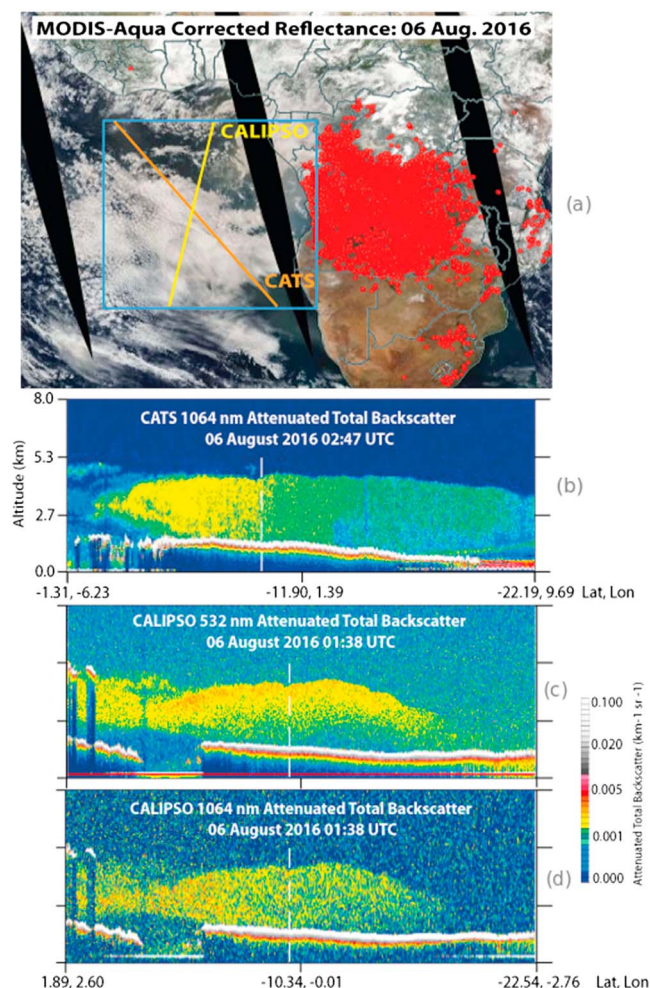


Figure 1. (a) A smoke above MBL cloud event on 6 August 2016. The red dots in the African Continent are fire events. Attenuated total backscatter of (b) CATS 1064 nm, (c) CALIPSO 532 nm, (c) and CALIPSO 1064 nm. The dashed lines correspond to the point where the CAT and CALIPSO tracks overlap with each other.

aerosols during the daytime in the CATS L2O Version 1-05 data products, though it performs well during nighttime observations. Future versions of CATS L2O data products will include layer detection at 60 km, but since Version 1-05 is used in this study, CATS daytime data were excluded.

For above-cloud aerosol (ACA), the more relevant difference between the algorithms is the preferred wavelength for atmospheric layer detection. The current CALIOP-SIBYL primarily uses 532 nm because it has higher signal-to-noise ratios and lower minimum detectable backscatter (MDB, weakest aerosol backscatter coefficient that can be detected) than the CALIOP 1064 nm data resulting in more accurate uniform cloud and aerosol layer detection [Vaughan *et al.*, 2009]. The CATS layer detection algorithm uses the 1064 nm attenuated scattering ratio because the CATS 532 nm data in Mode 7.2 are extremely noisy and the 1064 nm MDB is orders of magnitude lower [Yorks *et al.*, 2016]. For ACA detection specifically, the 1064 nm wavelength is preferred over the 532 nm wavelength for layer detection. The aerosol signal at 1064 nm has 16 times less molecular contamination compared to 532 nm. As discussed in section 1, the 532 nm backscatter signal may be insensitive to the entire vertical extent of absorbing aerosol layers. Because aerosol extinction is usually smaller at 1064 nm than 532 nm, and the CATS 1064 nm backscatter signal is very robust, the vertical extent of absorbing aerosol layers is fully captured from CATS 1064 nm backscatter profiles. It is worth mentioning that the current CATS operational algorithm uses $AB2CT < 360$ m as the threshold to detect the clouds embedded within aerosol layers (CEAL). When $AB2CT < 360$, the ACA and the cloud below are merged and identified a CEAL case.

2.2. CATS

CATS is an elastic backscatter lidar employing photon counting detection and two high-repetition rate lasers that operate at 532 and 1064 nm [McGill *et al.*, 2015] that has been operating on the International Space Station (ISS) since February 2015. The ISS orbit, which is at an altitude of ~ 415 km and a 51° inclination, allows CATS to observe locations at different local times each overpass (~ 60 days to complete full diurnal cycle) with roughly a 3 day repeat cycle.

The CATS layer detection algorithm is a threshold-based layer detection method that is nearly identical to the CALIOP-SIBYL technique with four distinct differences, namely, the use of 60 m vertical resolution, a single horizontal spatial resolution (5 km), the use of the 1064 nm wavelength rather than 532 nm, and a technique to identify clouds embedded within aerosol layers [Yorks *et al.*, 2015]. The CATS L2 Operational (L2O) CAD algorithm is a multidimensional PDF technique like the CALIOP one [Yorks *et al.*, 2015] but uses the layer-integrated attenuated backscatter at 1064 nm and other variables such as layer midtemperature and layer thickness instead of the layer-integrated backscatter color ratio due to the unreliable 532 nm data in Mode 7.2. The use of a single horizontal spatial resolution in the CATS algorithm misses optically thin cirrus clouds and

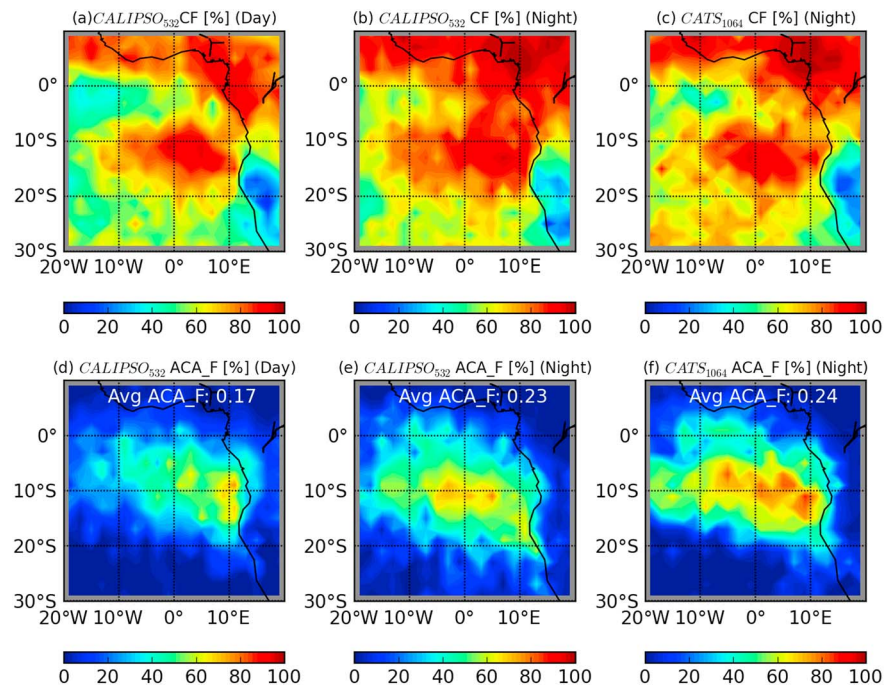


Figure 2. Multiyear (2015–2016) seasonal mean (July to October) cloud fraction in the SE Atlantic region based on (a) CALIPSO daytime, (b) CALIPSO nighttime, and (c) CATS nighttime observation. The seasonal mean ACA occurrence frequency from (d) CALIPSO daytime, (e) CALIPSO nighttime, and (f) CATS nighttime observations.

The detectability of the aerosol layer base using 532 and 1064 nm is demonstrated in Figure 1. CATS and CALIPSO passed over the same ACA layer over the SE Atlantic on 6 August 2016, although the differing orbits of the ISS and CALIPSO mean that the two curtains do not align exactly. There is a 0.1–1.0 km gap between cloud top and aerosol base in the attenuated total backscatter and vertical feature mask based on CALIOP 532 nm data. In contrast, CATS 1064 nm observation finds the aerosol plume to extend all the way to the cloud top, which is also confirmed by the CALIOP 1064 nm attenuated backscatter observation. The example clearly demonstrates the advantage of 1064 nm over 532 nm based layer detection technique for identifying the bottom of thick smoke layers. Although CALIOP also has the 1064 nm observation, it has not yet been utilized in the current operational algorithm. Note that the differences between CALIOP and CATS observations shown below are mainly due to the use of different wavelength (i.e., 532 nm versus 1064 nm) for layer detection. At the moment of writing, the CALIPSO operational product team is planning to make more use of the 1064 nm observations in their operational layer detection algorithm, which could significantly improve its retrievals for thick aerosol layers like the example in Figure 1.

3. Results

We have used the following criteria to identify ACA columns in both CALIOP and CATS layer products: (1) the cloud layer product identifies liquid phase cloud at the top layer of the profile, (2) the aerosol layer product identifies at least one layer of aerosol in the profile, and (3) the base height of at least one aerosol layer is higher than the top of the highest cloud layer. In the SE Atlantic region, most ACA cases are simple, with only one aerosol layer on top of single-layer MBL clouds. After the identification of ACA columns, we compute the AB2CT by calculating the difference between the minimum aerosol base height which is greater than maximum cloud top height and the maximum cloud top height. For CALIOP, we derived the ACA and cloud statistics for both daytime and nighttime conditions (though daytime and nighttime statistics are computed separately). The CATS results are only for nighttime since its aerosol retrieval does not perform well during daytime at the fixed 5 km horizontal resolution as discussed above.

Figures 2a–2c show the multiyear (2015–2016) SE Atlantic JASO Cloud Fraction (CF), defined as $CF = N_{cloudy}/N_{total}$ in $2^\circ \times 2^\circ$ grid boxes where N_{cloudy} is the number of cloudy columns and N_{total} is the number of total

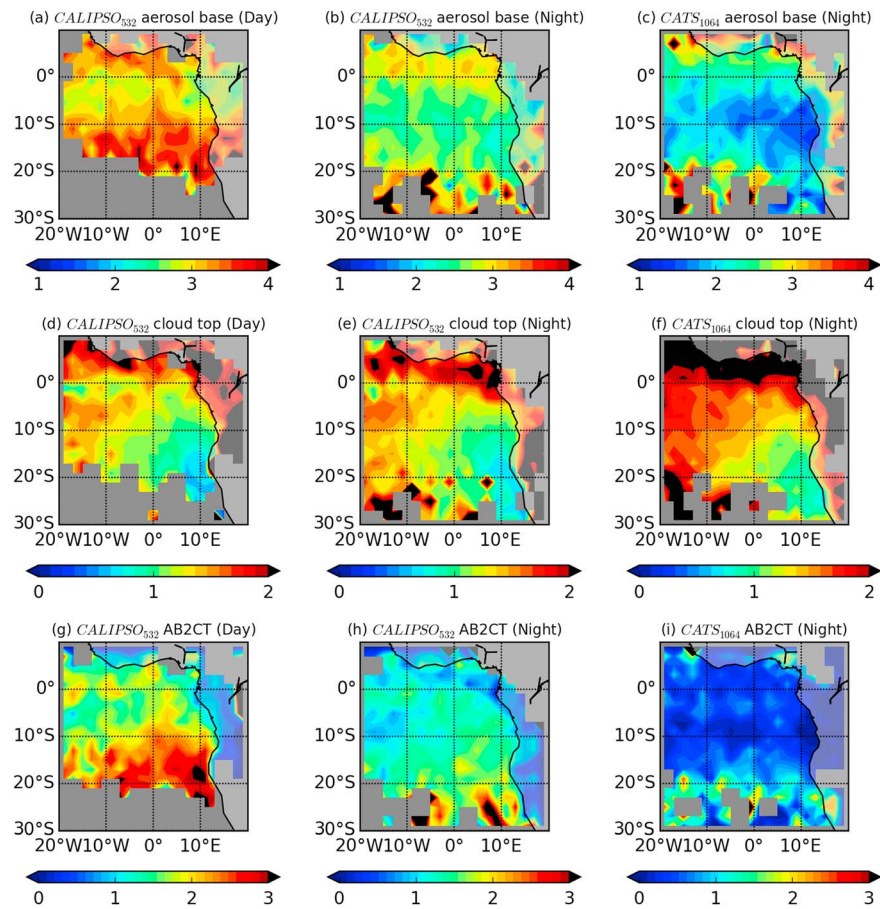


Figure 3. Multiyear (2015–2016) (top row) seasonal mean aerosol layer base height, (middle row) cloud layer top height, and (bottom row) aerosol base to cloud top (AB2CT) distance of ACA over the SE Atlantic region during JASO from CALIOP and CATS.

columns. Because we are interested in aerosol above low-level MBL clouds, ACA occurrence frequency (ACA_F) is shown in Figures 2d–2f which is defined as $ACA_F = N_{ACA}/N_{cloudy}$ where N_{ACA} is the number of ACA columns. Among the three data sets, CATS nighttime observations identify the highest ACA occurrence frequency, with domain averaged ACA_F around 0.24. CALIOP daytime observations have the lowest ACA occurrence frequency, with domain-averaged ACA_F only around 0.17. The CALIOP nighttime observations are comparable to the CATS nighttime observations (domain average $ACA_F \sim 0.23$). Some differences between the three data sets may have physical explanations. For example, CALIOP observes a larger CF during nighttime than during daytime, which is likely a result of the strong cloud diurnal cycle in the SE Atlantic region [Min and Zhang, 2014]. The other differences may stem from algorithm and instrument differences. For example, the lower ACA_F using daytime CALIOP might be an artifact due to the impact of background solar noise on the lidar retrieval [Liu et al., 2015].

Overall, the results in Figure 2 suggest that despite some minor differences, CALIOP and CATS observe similar geographical patterns of ACA in the SE Atlantic. We now focus on the vertical distribution of aerosol and cloud from the two instruments. Figure 3 shows the 2 year (2015–2016) mean aerosol layer base height (Figure 3, top row), cloud layer top height (Figure 3, middle row), and AB2CT distance (Figure 3, bottom row) of ACA over the SE Atlantic region during JASO from CALIOP and CATS. While the magnitudes differ, cloud top heights from all three data sets show a similar pattern, lowest off the coast of Namibia (near 20°S and 10°E) and gradually increasing along the northwest direction to about 2 km around 5°S and 15°W. In contrast to the similarity of cloud top height, the mean ACA base height from the three data sets show significant differences. ACA base height from daytime CALIOP observations is much higher than nighttime CALIOP, which is in turn higher than nighttime CATS. As a result, the AB2CT distance from nighttime

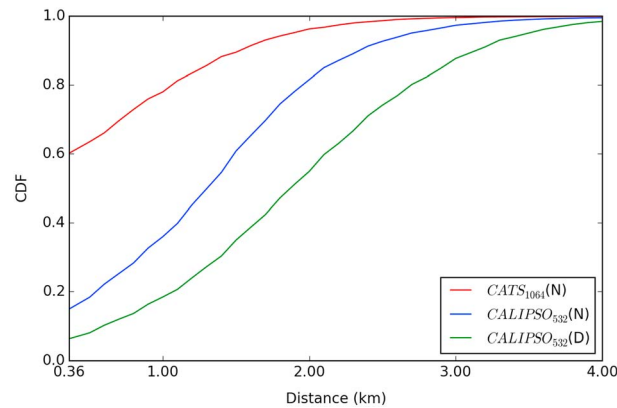


Figure 4. Cumulative probability distribution function (CDF) of the distance between aerosol layer bottom and cloud top (AB2CT distance). These curves are derived from the multiyear seasonal ACA data used in Figure 3.

60% of ACA cases are identified as CEAL (i.e., AB2CT < 360 m), in contrast to only 15% and 6% occurrence of such cases in CALIOP 532 nm nighttime (blue curve) and daytime (green curve) observations, respectively. Moreover, 82% and 64% of ACA cases have AB2CT > 1 km according to the daytime and nighttime CALIOP 532 nm observations, respectively, in contrast to 22% according to CATS observations.

Figure 5 shows meridionally averaged daytime (Figure 5a) and nighttime CALIOP (Figure 5b) 532 nm, and nighttime CATS 1064 nm (Figure 5c) observations of ACA top (dashed red line) and bottom (solid red line) height, cloud top height (blue line), and the fraction of ACA cases with AB2CT < 360 m (black line). Also

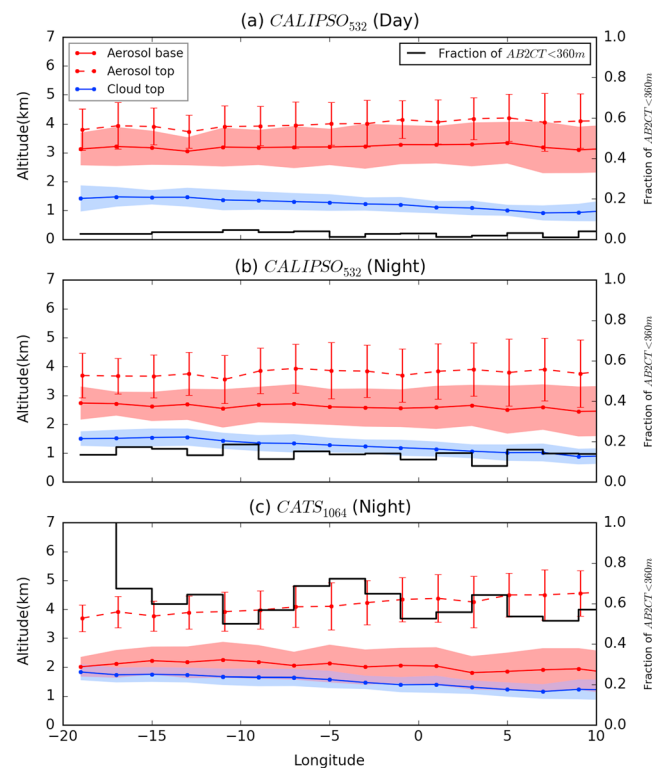


Figure 5. Meridionally averaged aerosol bottom (solid red line), top (dashed red line), and cloud top (solid blue line) heights, with fraction of AB2CT < 360 m (black line), for the SE Atlantic region during JASO 2015–2016. One standard deviation variability for each is denoted by the red error bars for aerosol top height, and by the red and blue shaded regions for the aerosol bottom and cloud top heights, respectively.

CATS is below 500 m in most of the SE Atlantic region, suggesting that the aerosol layer extends close to the cloud top. On the other hand, a clear separation between aerosol base and cloud top during both daytime and nighttime is implied by the CALIOP data, a likely result of the above mentioned CALIOP ACA layer detection issues.

We analyzed the AB2CT distances from the three observations further in Figure 4. Here we show the cumulative density function of the AB2CT distance for the sampling-masked ACA cases of Figure 3. According to CATS nighttime 1064 nm observations (red curve), about

60% of ACA cases are identified as CEAL (i.e., AB2CT < 360 m), in contrast to only 15% and 6% occurrence of such cases in CALIOP 532 nm nighttime (blue curve) and daytime (green curve) observations, respectively. Moreover, 82% and 64% of ACA cases have AB2CT > 1 km according to the daytime and nighttime CALIOP 532 nm observations, respectively, in contrast to 22% according to CATS observations.

shown are one standard deviation variability for ACA top (red error bars), ACA base (light red shades), and cloud top (light blue shades). All three observations show nearly the same top of aerosol layer, just below 4 km. The cloud top heights are also similar in all three observations, rising from 1 km near the coast westward to about 1.5–2.0 km at 19°W. Daytime CALIOP observes slightly higher cloud top height (domain average 1.39 km) compared to nighttime (domain average 1.33 km). Among all three data sets, the CATS detects the highest cloud top height (domain average 1.60 km). In contrast to aerosol top and cloud top heights, ACA base heights are substantially different among the three data sets. The CALIOP nighttime product (Figure 5b) gives domain-averaged ACA base height at 2.63 km; daytime CALIPSO retrievals (Figure 5a) are even higher. Nighttime CATS 1064 nm (Figure 5c), however, observes a significantly lower ACA domain-averaged base height around 2 km.

Even after considering one standard deviation variability, there is still a clear separation between the ACA base and cloud top in both the daytime

(Figure 5a) and nighttime (Figure 5b) CALIOP retrievals, confirmed by the small values of the fraction of AB2CT < 360 m throughout the domain. With CATS (Figure 5c), however, there is clear evidence that the ACA base and cloud top are in much closer proximity than is implied by CALIOP 532 nm observation, as the fraction of AB2CT < 360 m is mostly around 60%.

4. Summary and Discussion

The microphysical indirect effects of the seasonally transported aerosols in the SE Atlantic are often overlooked in the literature. This is partly because CALIOP's 532 nm based operational layer detection algorithm often detects the aerosol layer bottom too high and thereby suggests that the above-cloud aerosol layer is well separated from the underlying clouds. The newly launched CATS mission provides a new data set of the vertical distribution of aerosol and clouds. Several instrument and algorithm advantages of CATS, chiefly among which is the primary use of 1064 nm for layer detection, allow it to better identify the full vertical extent of the SE Atlantic ACA layer than CALIOP 532 nm product. We have compared the current CATS and CALIPSO products during JASO of 2015 and 2016 over the SE Atlantic. The CF, ACA_F, and cloud top geographical patterns from the two instruments agree well. However, CATS 1064 nm observes the ACA layer bottom height much lower and much closer to the underlying cloud top than CALIOP 532 nm does. According to CATS, about 60% of the ACA cases have an AB2CT < 360 m, in contrast to the 15% and 6% based on CALIOP nighttime and daytime 532 nm observations, respectively.

Our study provides direct evidence that space-based lidar layer detection at 1064 nm is more representative of the true ACA scene compared to 532 nm. More importantly, our study suggests that the occurrence of aerosol entrainment into clouds might be much more frequent than previously thought based on CALIOP 532 nm observations. This implies that the microphysical indirect effects could be an important mechanism through which the transported aerosol influences the clouds and radiation in SE Atlantic region. Finally, an accurate measurement of the vertical distribution of aerosols would also help us better understand the semi-direct effects of the smoke aerosols.

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